

Roll No.

TCS-523

B. TECH. (CSE) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022
BIG DATA AND MACHINE LEARNING USING PYTHON

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Differentiate between Supervised, Unsupervised and Semi-supervised machine learning. 10 Marks (CO1)

OR

- (b) Discuss any *four* examples of machine learning applications in brief. 10 Marks (CO1)

2. (a) Explain Regression and Classification with an example. 10 Marks (CO1)

OR

- (b) Explain Linear Regression with example. 10 Marks (CO1)

3. (a) How does K fold cross validation work ? Explain with an example. 10 Marks (CO2)

OR

- (b) Explain Naïve Bayes Classifier with an example. 10 Marks (CO2)

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4. (a) What is a decision tree. Discuss the use of decision tree for classification purpose with an example. 10 Marks (CO2)

OR

- (b) What are the different methods for measuring classifier performance ?

10 Marks (CO2)

5. (a) Explain, how Support Vector Machine can be used for classification of linearly separable data. 10 Marks (CO1)

OR

- (b) Explain the following :

10 Marks (CO1)

- (i) Gradient descent
- (ii) Learning rate
- (iii) Mean squared error cost function